BLUEPRINT OVERVIEW

Top Five Areas

	TODAY	TARGET	AGENTIC AI
Reduce cost	10p+ per assessment Cold container per request Economics break at scale	<1p per assessment Warm inference pool (AKS) On-device signal extraction	Cost thresholds monitored via Grafana alerting Ops agent correlates spikes with known change
Improve performance	No latency guarantee, users wait for model All compute in cloud Video transmitted from device	Async queue, warm pool Horizontal autoscaling Signal extracted on-device Metal/Vulkan GPU pipeline	Grafana alerting detects performance degradation ML Validation agent monitors output for model drift
Release safely at pace	Manual release gates No formal API contract Test evidence ad hoc	Automated pipeline Immutable artefacts Every sprint a release candidate	Compliance agent gates PRs Code Review agent on every PR Doc agent drafts changelogs
Observability	Limited visibility into production failures No mobile SNR telemetry No root cause analysis of failure	Grafana + Prometheus for metrics Loki for logs Sentry for error tracking Signal quality reported per assessment	Ops agent reads metrics Creates GitHub Issues from alerts AI surfaces, humans decide
Governance & planning	Decisions undocumented Sprint cadence informal Traceability manual	ADRs before code 1-week sprints, DoD with regulatory gates eQMS integrating with development process	Compliance agent enforces traceability at PR time Governance visible continuously

Reduce Cost

SUMMARY

Current architecture: A new container is created per assessment to guarantee patient data isolation, a safe pragmatic call, but expensive. Each cold start loads the ML model from scratch.
At low volumes this is manageable, it becomes a commercial problem as we scale.

TECH STACK

Current - Flask
Target - AKS · Azure Service Bus · FastAPI

APPROACH & RECOMMENDATION

Refactor to a stateless inference service enabling a warm pool where model weights stay loaded in memory. Patient isolation is guaranteed by design, not container boundary. Async queue (Azure Service Bus) absorbs demand spikes.

On-device signal extraction moves face detection, ROI extraction, and POS + CHROM processing to the device. Only the processed BVP signal is transmitted. Removes a further chunk of cloud compute per assessment.

ALTERNATIVES CONSIDERED & REJECTED

Serverless inference (Azure Functions) the cold start still happens, it’s just hidden. Full stateless rewrite is the right destination but too risky to do in one go on a live clinical system. Incremental refactor is safer.